

The 3R planar robot in Fig. 1 is commanded at the joint velocity level. The robot has to perform two tasks simultaneously, if possible. The first task is to keep the second link vertical and upwards at any time. The second task is to follow a desired cyclic Cartesian trajectory  $p_d(t) \in \mathbb{R}^2, t \in [0, T]$ , for the end-effector position. Provide the actual expressions of all terms in a task priority control law, with the given order of tasks. Determine the robot configurations for which both tasks can be perfectly executed together, and define accordingly the region of the plane where this can happen. Which would be the control law in this case? With link lengths  $L_1 = L_2 = L_3 = 0.5$  [m], compute the numerical value of  $\dot{q} \in \mathbb{R}^3$  using the task priority law at  $q_0 = (0 \ \pi/2 \ -\pi/2)^T$  [rad] for  $\dot{p}_d = (0.1 \ -0.5)^T$  [m/s]. Finally, when errors are present during the execution of these tasks, how should the control law be modified in order to reduce them?

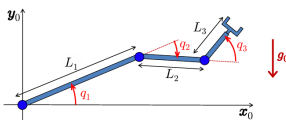


Figure 1: A 3R planar robot with its joint variables and generic link lengths.

the first task is  $\zeta_1 = q_1 + q_2$  with  $\zeta_{1d} = \frac{\pi}{2} \Rightarrow \dot{\zeta}_{1d} = 0$

then  $\zeta_2 = \begin{cases} L_1 c_1 + L_2 c_{12} + L_3 c_{123} \\ L_1 s_1 + L_2 s_{12} + L_3 s_{123} \end{cases}$  with  $\zeta_{2d} = p_d \Rightarrow \dot{\zeta}_{2d} = \dot{p}_d$  we compute the two Jacobians

$$J_1 = \frac{\partial \zeta_1}{\partial q} = \begin{pmatrix} 1 & 1 & 0 \end{pmatrix} \quad J_2 = \frac{\partial \zeta_2}{\partial q} = \begin{pmatrix} -L_1 s_1 - L_2 s_{12} - L_3 s_{123} & -L_2 s_{12} - L_3 s_{123} & -L_3 s_{123} \\ L_1 c_1 + L_2 c_{12} + L_3 c_{123} & L_2 c_{12} + L_3 c_{123} & L_3 c_{123} \end{pmatrix}$$

task 1)  $\dot{q}_1 = J_1^\# \dot{\zeta}_{1d} = \begin{pmatrix} 0 & 0 & 0 \end{pmatrix}^T$

task 2)  $P_1 = I - J_1^\# J_1 = \begin{pmatrix} 1/2 & -1/2 & 0 \\ -1/2 & 1/2 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad J_2 P_1 = \begin{pmatrix} -\frac{1}{2} L_1 s_1 & \frac{1}{2} L_1 s_1 & -L_3 s_{123} \\ \frac{1}{2} L_1 c_1 & -\frac{1}{2} L_1 c_1 & L_3 c_{123} \end{pmatrix}$

the sol is  $\dot{q}_2 = (J_2 P_1)^\# \dot{p}_d$ .

the two tasks can be executed together perfectly if  $J_A = \begin{pmatrix} J_1 \\ J_2 \end{pmatrix}$  is

non singular:

we impose  $s_{12} = 1 \quad c_{12} = 0$   
 $q_1 + q_2 = \pi/2 \Rightarrow s_{123} = c_3 \quad c_{123} = -s_3$

$$J_A = \begin{pmatrix} 1 & 1 & 0 \\ -L_1 s_1 - L_2 s_{12} - L_3 s_{123} & -L_2 s_{12} - L_3 s_{123} & -L_3 s_{123} \\ L_1 c_1 + L_2 c_{12} + L_3 c_{123} & L_2 c_{12} + L_3 c_{123} & L_3 c_{123} \end{pmatrix}$$

so  $J_A$  becomes

for sure is singular if  $q_1, q_3 \in \{0, \pi\}$

$$J_A = \begin{pmatrix} 1 & 1 & 0 \\ -L_1 s_1 - L_2 - L_3 c_3 & -L_2 - L_3 c_3 & -L_3 c_3 \\ L_1 c_1 - L_3 s_3 & -L_3 s_3 & -L_3 s_3 \end{pmatrix}, \quad J_2^\star = \begin{pmatrix} 1 & 0 & 0 \\ -L_2 - L_3 & -L_2 - L_3 & -L_3 \\ L_1 & 0 & 0 \end{pmatrix}$$

columns 2 and 3 becomes lin. dependent.

$$J_2 P_1(q_0) = \begin{pmatrix} 0 & 0 & 0 \\ \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \end{pmatrix} \Rightarrow \dot{q} = \begin{pmatrix} 0 & \frac{1}{4} \\ 0 & -\frac{1}{4} \\ 0 & \frac{1}{2} \end{pmatrix} \begin{pmatrix} 0.1 \\ -0.5 \end{pmatrix} = \begin{pmatrix} -\frac{1}{8} & \frac{1}{8} & \frac{1}{4} \end{pmatrix}^T$$

The RP planar robot shown in Fig. 2 lives in a vertical plane and may collide with some (human) obstacle when in motion. Its controller is therefore equipped with a momentum-based collision detection algorithm that generates a residual vector  $r \in \mathbb{R}^2$  as monitoring signal. Provide the explicit symbolic expressions of the two scalar components of  $r$  (introduce the needed kinematic and/or dynamic quantities). Suppose that, at time  $t = t_c$ , a collision occurs on the robot tip with an impact force  $F_c$  that is purely normal to the second link. What will be the instantaneous value of the time derivative of the residual vector  $\dot{r}(t_c)$ ?

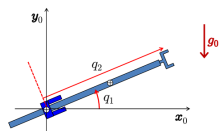


Figure 2: A RP robot moving in a vertical plane.

the CoM of the first link is always in the origin, hence its kinetic energy

is  $T_1 = \frac{1}{2} I_1 \dot{q}_1^2$ . The CoM of the second link is:

$$p_{c2} = \begin{cases} (q_2 - d) c_1 \\ (q_2 - d) s_1 \end{cases} \quad \text{where } d \text{ is the distance between the gripper and the CoM.}$$

$$v_{c2} = \begin{cases} \dot{q}_2 c_1 - (q_2 - d) s_1 \dot{q}_1 \\ \dot{q}_2 s_1 + (q_2 - d) c_1 \dot{q}_1 \end{cases} \quad \text{the angular vel. is just } \omega_2 = \dot{q}_1$$

$$v_{c2}^T v_{c2} = \dot{q}_2^2 + (q_2 - d)^2 \dot{q}_1^2 = \dot{q}_2^2 + \dot{q}_1^2 (q_2^2 + d^2 - 2q_2 d)$$

$$\Rightarrow T_2 = \frac{1}{2} m_2 (\dot{q}_2^2 + \dot{q}_1^2 (q_2^2 + d^2 - 2q_2 d)) + \frac{1}{2} I_2 \dot{q}_1^2 \quad \text{so: } \begin{cases} e_1: I_1 + I_2 + m_2 d^2 \\ e_2: -2m_2 d \\ e_3: m_2 \end{cases}$$

$$M_{11} = I_1 + I_2 + m_2 d^2 + m_2 q_2^2 - 2m_2 d q_2 = e_1 + e_2 q_2 + e_3 q_2^2$$

$$M_{22} = m_2 = e_3$$

$$M = \begin{pmatrix} e_1 + e_2 q_2 + e_3 q_2^2 & 0 \\ 0 & e_3 \end{pmatrix}$$

$$\text{the pot. energy is } U = m_2 g_0 (q_2 - d) s_1 \Rightarrow g = \begin{cases} m_2 g_0 (q_2 - d) c_1 \\ m_2 g_0 s_1 \end{cases}$$

$$C_1 = \frac{1}{2} \left( \frac{\partial M_1}{\partial q} + \left( \frac{\partial M_1}{\partial \dot{q}} \right)^T - \frac{\partial M}{\partial q_1} \right) = \frac{1}{2} e_2 + e_3 q_2 \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \Rightarrow c_1 = (e_2 + 2e_3 q_2) \dot{q}_1 \dot{q}_2$$

$$C_2 = \frac{1}{2} \left( \frac{\partial M_2}{\partial \dot{q}} + \left( \frac{\partial M_2}{\partial \dot{q}} \right)^T - \frac{\partial M}{\partial \dot{q}_2} \right) = \begin{pmatrix} -\frac{1}{2} e_2 - e_3 q_2 & 0 \\ 0 & 0 \end{pmatrix} \Rightarrow c_2 = -\dot{q}_1^2 \left( \frac{1}{2} e_2 - e_3 q_2 \right)$$

$$r = K(p - \int (z + c - y + r) dt - p(0))$$

before any collision,  $r = 0$

$$K > 0 \quad p = M \dot{q} \quad \begin{pmatrix} J_1 & J_2 \end{pmatrix} \begin{pmatrix} \kappa_1 \\ \kappa_2 \end{pmatrix}^* \quad \text{DK} \quad \rightarrow J_1 q_2 c_1 + J_2 q_2 s_1 = 0$$

$$\text{In } t_c \text{ we have } F_c \text{ s.t. } F_c^T K(q) = 0 \quad \text{where } K(q) = \begin{cases} q_2 c_1 \\ q_2 s_1 \end{cases} \quad \begin{matrix} q_2 (s_1 c_1 + s_2 s_1) = 0 \\ \Rightarrow s_1 c_1 + s_2 s_1 = 0 \end{matrix}$$

$$\tau_c = J^T F_c = \begin{pmatrix} -q_2 s_1 & q_2 c_1 \\ c_1 & s_1 \end{pmatrix} F_c \Rightarrow \dot{r} = K J^T F_c =$$

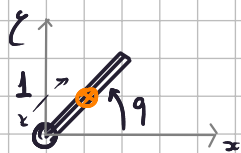
$$K \begin{pmatrix} -q_2 s_1 s_1 + q_2 c_1 s_2 \\ c_1 s_1 + s_1 s_2 \end{pmatrix}^* \Rightarrow \begin{pmatrix} \kappa_1 q_2 (c_1 s_2 - s_1 s_1) \\ 0 \end{pmatrix}$$

no torque error on joint 2: we can't say that collision happened on link 2.

### Exercise 3

An actuated pendulum under gravity should perform a rest-to-rest swing-up maneuver from the downward position  $\theta(0) = 0$  to the upward position  $\theta(T) = \pi$  in a total time  $T$ , using a bang-coast-bang acceleration profile with symmetric acceleration and deceleration phases, each of duration  $T_s = T/4$ . The link of the pendulum is a thin rod of length  $l = 2$  [m], with uniformly distributed mass  $m = 10$  [kg] and baricentral inertia  $I_c = ml^2/12$  [kg·m<sup>2</sup>]. The motor at the link base can deliver a maximum absolute torque  $\tau_{max} = 200$  [Nm]. Determine the minimum time  $T_{min}$  in the chosen class of trajectories such that the motion is feasible. Sketch the resulting angular position, velocity, acceleration, and torque profiles.

We decide to choose  $q$  as the angle between the rod and the  $x$ -axis so the displacement is from  $-\pi/2$  to  $\pi/2$ .



The position of the CoM is  $p = (c \ s)^T$  so  $\dot{p} = (-s \dot{q} \ c \dot{q})^T$

$$\text{then } \dot{p}^T \dot{p} = \dot{q}^2 \Rightarrow T = \frac{1}{2} m \dot{q}^2 + \frac{1}{2} I \dot{q}^2 \Rightarrow e_1 = m + I = 40/3 \approx 13.3$$

The potential energy is  $U = m g_0 \sin q \Rightarrow g = \frac{\partial U}{\partial q} = \overbrace{m g_0 \cos q}^{e_2 \approx 98} = e_2 \cos q$

The dyn. model is  $e_1 \ddot{q} + e_2 \cos q = \tau$ , to control  $\ddot{q}$  we apply FBL:

$$\tau = e_2 \cos q + e_1 a \Rightarrow \ddot{q} = a. \text{ We need an upper bound for } a:$$

$$|\tau| = |e_2 + e_1 a| \leq 200 \Rightarrow |a| \leq \frac{1}{e_1} (200 - e_2) = A \approx 7.65$$

max speed  
↓

total displ.  
↓

$$V = A \cdot \frac{T}{4} \Rightarrow S = V \cdot \frac{T}{2} + V \cdot \frac{T}{4} = VT \frac{3}{4} = AT^2 \frac{3}{16} = \frac{459}{320} T^2 = \pi \Rightarrow T \approx 1.48 \text{ sec}$$